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Multi-Horizon Hazard Models for Battery Failure Prediction: Within-Dataset Reliability, Cross-Chemistry Transferability, and Calibration Analysis

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***Abstract*— This paper presents a multi-horizon hazard prediction framework for predicting lithium-ion battery failure that integrates probability calibration, cross-chemistry evaluation, and statistical significance testing. Three tree ensemble models (XGBoost, LightGBM, Random Forest) with 300 estimators and a GRU baseline are trained using the NASA and CALCE LCO datasets and evaluated on Oxford and MIT-Stanford Severson LFP datasets across four horizons (10, 20, 30, 50 cycles). A fairness-based comparison of Platt sigmoid and isotonic calibration is combined with an SOH ablation study and DeLong significance testing. The key finding is that apparent cross-chemistry transfer success depends largely on the State-of-Health (SOH) feature rather than transferable degradation patterns: removing SOH causes AUC to drop from 0.890 to 0.413 for XGBoost in the NASA+CALCE → Oxford transfer ($p = 7.2 \times 10^{-51}$). For within-dataset evaluation, Platt-calibrated models achieve AUC ≥ 0.85 in 30 of 32 model-horizon-dataset combinations, and Platt sigmoid consistently outperforms isotonic regression.**

***Index Terms*— Battery management systems, machine learning, probability calibration, prognostics and health management, transfer learning.**

I. Introduction and Related Work

Lithium-ion batteries are fundamental to electric vehicles and grid-scale energy storage because of their high energy density, fast charging capability, and long cycle life. However, their degradation is highly nonlinear, governed by complex electrochemical processes, making battery failure prediction challenging. Although failures are relatively infrequent, they can cause significant safety hazards and economic losses, motivating reliable multi-horizon prognostics for advanced battery management systems [1]. Battery prognostics has primarily focused on state of health (SOH) estimation and remaining useful life (RUL) prediction. Berecibar et al. [1] reviewed conventional SOH methods, including coulomb counting, impedance spectroscopy, Kalman filtering, fuzzy logic, and machine learning. Recent studies have adopted deep learning, including STL-LSTM with spatio-temporal attention [2], multilayer perceptrons [3], and SHAP-assisted feature engineering [4]. Similarly, extensive reviews of RUL prediction are available in [5]–[7], while LightGBM [8], GRU-based [9], hybrid neural networks [10], and convolutional neural networks [11] have demonstrated competitive lifetime prediction performance. Survival analysis has recently emerged as an alternative paradigm for battery reliability prediction. Ibraheem et al. [12] employed Cox proportional hazards modeling, Li et al. [13] proposed early-cycle feature-based prediction, and Yao et al. [14] developed a physics-guided machine learning framework for capacity degradation. Cross-chemistry transfer has not received much direct attention. Pang et al. [15] proposed a Transformer-BiLSTM framework for multi-chemistry SOH estimation, though their focus was SOH rather than failure prediction. Probability calibration is similarly underexplored, despite established methods such as Platt scaling [16], isotonic regression [17], and survival calibration [18]. Model interpretability using SHAP [19] and public benchmark datasets [20], [21] continue to advance the field.

Nevertheless, a clear research gap remains. Most studies evaluate a single dataset, chemistry, horizon, or metric, while cross-chemistry transfer, calibration, and significance testing are rarely studied together. It is also unclear whether transfer performance reflects genuine degradation patterns or chemistry-specific features, a distinction our ablation study is designed to test. The DeLong test [22] is common for ROC comparison but rarely applied in battery prognostics, and while the Severson dataset [23] enables large-scale LFP validation, it has not been widely paired with cross-chemistry experiments.

This paper addresses that gap. We train XGBoost, LightGBM, Random Forest, and a GRU baseline on the NASA and CALCE LCO datasets, then evaluate on the Oxford and Severson LFP datasets across four horizons (10, 20, 30, 50 cycles). We compare Platt and isotonic calibration under a shared protocol, apply DeLong significance testing, and run an SOH ablation to test what is actually transferring. The key finding is that: apparent cross-chemistry transfer success depends largely on SOH acting as a chemistry-specific signal, not a shared degradation pattern.

The main contributions of this work are as follows:

1. A multi-chemistry extension of the multi-horizon hazard prediction framework evaluated on four public datasets covering LCO and LFP batteries.
2. A fair comparison of Platt sigmoid calibration and isotonic regression for battery failure prediction.
3. An SOH ablation study supported by the DeLong significance test to examine cross-chemistry transfer.

The remainder of this paper is organized as follows. Section II presents the proposed methodology. Section III reports the experimental results and discussion. Section IV concludes the paper, including its limitations and directions for future work.

II. METHODOLOGY

The proposed framework integrates multi-horizon hazard prediction, probability calibration, cross-chemistry transfer evaluation, and statistical significance testing into a unified battery failure prediction workflow (Fig. 1).

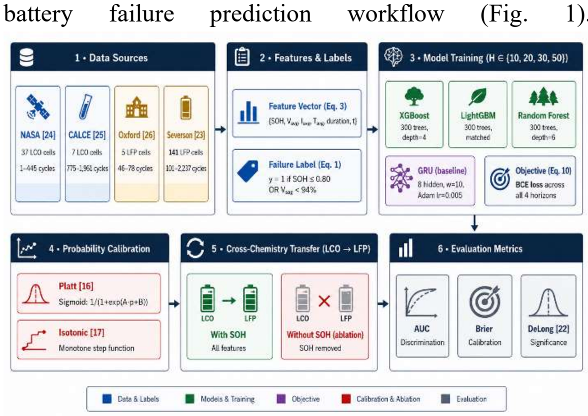


Fig. 1. Overview of the proposed multi-horizon battery failure prediction framework

A. Datasets and Feature Representation

Four public datasets spanning two battery chemistries are used (Table I). The NASA PCoE [24] and CALCE CX2 [25] datasets are used for training, while the Oxford Battery Degradation Dataset [26] and MIT-Stanford Severson dataset [23] serve as cross-chemistry transfer targets. A battery is labeled as failed when SOH falls below 0.80 or the average voltage sag drops below 94% of the first ten-cycle baseline.

TABLE I. SUMMARY OF THE FOUR BATTERY DATASETS.

Dataset	Cells	Chem.	Cycles/Cell	Role
NASA 18650 [24]	37	LCO	1–445	Training
CALCE CX2 [25]	7	LCO	775–1,961	Training
Oxford LFP [26]	5	LFP	46–78	Transfer target
Severson [23]	141	LFP	101–2,237	Transfer target

The Oxford cycle index is derived from the original MATLAB structure (up to 8,200), whereas the reported range (46–78) corresponds to valid discharge cycles after preprocessing. Cycle ranges for all datasets refer to the cleaned data used in experiments.

The feature vector consists of standard Battery Management System (BMS) measurements requiring no additional sensors:

$$x_t = \{\mathrm{SOH}_t, V_{\mathrm{avg}}, I_{\mathrm{avg}}, T_{\mathrm{avg}}, \mathrm{duration}, t\} \tag{1}$$

Tree-ensemble models use x_t directly, whereas the GRU processes a rolling window of $w = 10$ cycles:

$$X_t = [x_{t-w}, \dots, x_t] \tag{2}$$

In the CALCE dataset, avg_temp and duration are completely missing. Tree ensembles retain these variables using constant-zero imputation (preventing informative splits), whereas the GRU excludes them and uses only cycle, average voltage, minimum voltage, and SOH during cross-chemistry experiments.

B. Operational Reliability Formulation

Rather than estimate SOH or remaining useful life directly, we predict failure probability within a fixed operating horizon.

$$Y_{t,H} = \begin{cases} 1, & \text{if failure occurs within } (t, t+H] \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where Hdenotes the prediction horizon. The conditional failure probability is

$$P_{\mathrm{fail}}(t,H) = P\big(T_f \le t + H \mid T_f > t, x_t\big) \tag{4}$$

where Tf is the failure cycle. This framing treats battery prognostics as a discrete-time survival problem, aligned with how a BMS would use the output operationally.

C. Multihorizon Discrete-Time Hazard Learning

We predict failure probability at four operating horizons:

$$H \in \{10, 20, 30, 50\} \tag{5}$$

For each horizon, the discrete-time hazard is

$$h(t,H) = P(T_f = t + H \mid T_f > t, X_t) \tag{6}$$

with survival probability

$$S(t,H) = \prod_{k=1}^{H} (1 - h(t,k)) \tag{7}$$

and cumulative failure probability

$$P_{\text{fail}}(t,H) = 1 - S(t,H) \tag{8}$$

The predictive model learns

$$f_\theta : X_t \to P_{\text{fail}}(t,H) \tag{9}$$

by jointly optimizing all horizons using binary cross-entropy:

$$L = -\sum_{t,H} \left[y_{t,H} \log P_{\text{fail}} + \left(1 - y_{t,H}\right) \log(1 - P_{\text{fail}}) \right] \tag{10}$$

We evaluate four classifiers: XGBoost (max_depth=4, learning rate=0.05, subsample=0.8, colsample=0.8, min_child_weight=5), LightGBM with matched hyperparameters (min_child_samples=20), Random Forest (300 trees, max_depth=6), and a single-layer GRU (8 hidden units, 10-cycle input window, BCEWithLogits loss, Adam at learning rate=0.005, early stopping with patience=10, up to 50 epochs). All tree-ensemble models were configured with 300 estimators to keep the comparison controlled.

D. Probability Calibration

The raw predictions are calibrated using Platt scaling [16] and isotonic regression [17] under an identical training-fold protocol. Platt scaling fits a logistic model,

$$\hat{P}_{\text{cal}} = g_P(\hat{P}_{\text{fail}}) = \frac{1}{1 + \exp(A\hat{P}_{\text{fail}} + B)} \tag{11}$$

where $A$and $B$are estimated by maximum likelihood. Isotonic regression learns a monotonic mapping,

$$\hat{P}_{\text{cal}} = g_{\text{iso}}(\hat{P}_{\text{fail}}) \tag{12}$$

We fit both calibrators directly on training-fold scores rather than using CalibratedClassifierCV, which implicitly ensembles across folds and would confound the calibration comparison.

Calibration quality is assessed by

$$P\big(\text{ failure } \mid \hat{P}_{\text{cal}} = p\ \big) \approx p \tag{13}$$

E. Cross-Chemistry Transfer Protocol

Models are trained on NASA, CALCE, or their combination and evaluated on Oxford and Severson, with results averaged across test cells. We consider two feature settings, the full feature set and SOH removed, to test whether cross-chemistry performance reflects transferable degradation or mainly depends on SOH. Significance is assessed using the DeLong test [22].

III. Results and Discussion

A. Within-Dataset Reliability

The framework performed well on both NASA and CALCE. Table II shows Platt-calibrated AUC of at least 0.85 in 30 of 32 model-horizon combinations, with the exceptions being GRU on CALCE at H=20 (0.779) and H=30 (0.768). Mean AUC ranged from 0.844 (GRU, CALCE) to 0.913 (Random Forest, NASA), and Brier scores stayed consistently low, indicating reliable probability estimates. The GRU showed the most variation across folds, while the tree ensembles stayed more stable, with AUC differences across horizons generally under 0.05, as shown in Fig. 2.

TABLE II. WITHIN-DATASET PLATT-CALIBRATED AUC AND BRIER SCORE PER HORIZON.

Model	**Dataset**	**H=10**	**H=20**	**H=30**	**H=50**	**Mean**	**Brier**
GRU	CALCE	0.944	0.779	0.768	0.885	0.844	0.124
GRU	NASA	0.930	0.884	0.876	0.868	0.889	0.191
LightGBM	CALCE	0.908	0.924	0.886	0.916	0.909	0.101
LightGBM	NASA	0.912	0.909	0.892	0.901	0.903	0.250
Random Forest	CALCE	0.895	0.889	0.884	0.864	0.883	0.095
Random Forest	NASA	0.911	0.901	0.912	0.927	0.913	0.209
XGBoost	CALCE	0.916	0.926	0.911	0.899	0.913	0.101
XGBoost	NASA	0.903	0.899	0.910	0.918	0.907	0.197

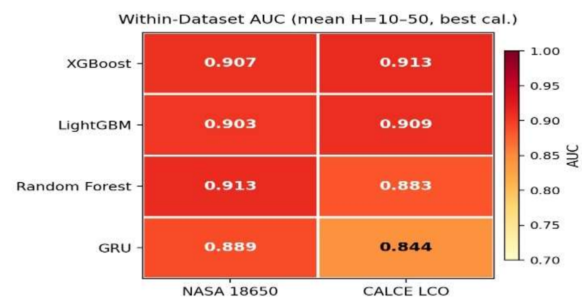


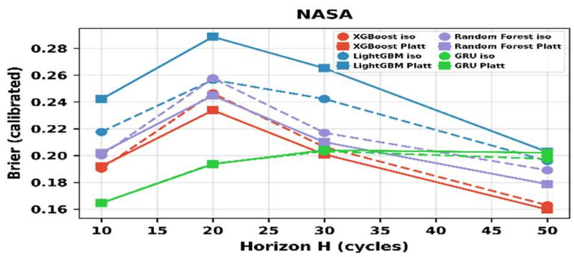
Fig. 2. Average AUC across prediction horizons for the NASA and CALCE datasets.

B. Calibration Comparison: Platt Outperforms Isotonic

Table III compares Platt sigmoid calibration with isotonic regression. Platt calibration consistently reached a higher mean AUC on both datasets, with the largest improvement was observed on CALCE (0.902 versus 0.692). Despite that gap, both methods produced similar Brier scores, so probability calibration quality was roughly comparable between them. On the NASA dataset, isotonic regression achieved a slightly lower mean Brier score (0.215 versus 0.219); however, the substantially higher AUC of Platt calibration on both datasets (0.902 vs. 0.692 on CALCE; 0.908 vs. 0.834 on NASA) indicates that it provides superior discrimination while preserving comparable calibration. Overall, Platt calibration provides a better balance between discrimination and calibration, making it the preferred method for the subsequent experiments. Figure 3(a) and (b) compare the calibrated Brier scores of Platt scaling and isotonic regression across prediction horizons for the NASA and CALCE datasets, respectively.

TABLE III. CALIBRATION METHOD COMPARISON (MEAN ACROSS TREE MODELS)

Dataset	Iso. AUC	Platt AUC	Iso. Brier	Platt Brier
CALCE	0.692	0.902	0.099	0.099
NASA	0.834	0.908	0.215	0.219



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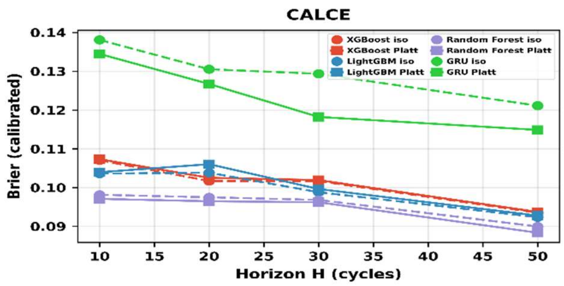


(b)

Fig. 3. Calibrated Brier score across prediction horizons for Platt and isotonic calibration using (a) NASA and (b) CALCE datasets.

C. Cross-Chemistry Transfer and Statistical Significance

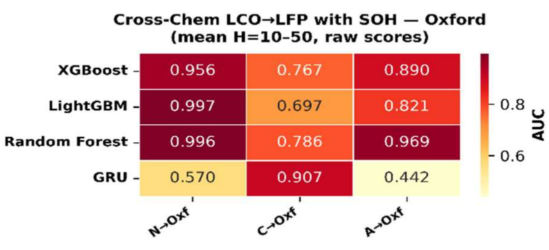
Because Platt and isotonic calibration parameters are chemistry-specific, cross-chemistry transfer is reported using raw AUC. Table IV compares transfer performance with and without the SOH feature across all training-target-model combinations, and Fig. 4 and Fig. 5 visualize the same results with (a) Oxford and (b) Severson as targets. When SOH was included, tree-ensemble models achieved raw AUC values mostly between 0.85 and 1.00: for the combined NASA+CALCE → Oxford experiment, XGBoost, LightGBM, and Random Forest reached 0.890, 0.821, and 0.969, respectively, rising to 0.995, 0.988, and 0.998 on Severson. Removing SOH caused a sharp decline for every model, the AUC of XGBoost decreased from 0.890 to 0.413 on NASA+CALCE → Oxford, while the GRU was the least stable across chemistries (raw AUC 0.055 to 0.500). Table V confirms these drops are statistically significant: for the Oxford ablations, XGBoost, LightGBM, and Random Forest returned p-values of 7.2×10^{-51}, 2.6×10^{-64}, and 6.5×10^{-36}, and all Severson ablation p-values fell below floating-point precision. Within-dataset model comparisons showed smaller, only occasionally significant differences. Together, these findings suggest that apparent cross-chemistry transfer success depends largely on SOH as a chemistry-specific indicator rather than a genuinely transferable degradation signal.

TABLE IV. CROSS-CHEMISTRY TRANSFER PERFORMANCE WITH AND WITHOUT SOH.

Training	Model	Target	Raw AUC (+/−SOH)	Iso AUC (+/−SOH)
CALCE	GRU	Oxford	0.907 / 0.055	0.500 / 0.500
		Severson	0.481 / 0.487	0.500 / 0.500
	LightGBM	Oxford	0.697 / 0.400	0.508 / 0.508
		Severson	0.861 / 0.801	0.608 / 0.599
	Random Forest	Oxford	0.786 / 0.484	0.513 / 0.510
		Severson	0.985 / 0.837	0.678 / 0.615
	XGBoost	Oxford	0.767 / 0.373	0.509 / 0.513
		Severson	0.969 / 0.806	0.609 / 0.604
NASA+CALCE	GRU	Oxford	0.442 / 0.155	0.500 / 0.500
		Severson	0.494 / 0.500	0.503 / 0.495
	LightGBM	Oxford	0.821 / 0.423	0.506 / 0.512
		Severson	0.988 / 0.828	0.642 / 0.603
	Random Forest	Oxford	0.969 / 0.497	0.596 / 0.541
		Severson	0.998 / 0.872	0.957 / 0.838
	XGBoost	Oxford	0.890 / 0.413	0.510 / 0.510
		Severson	0.995 / 0.853	0.702 / 0.653
NASA	GRU	Oxford	0.570 / 0.458	0.586 / 0.436
		Severson	0.573 / 0.496	0.499 / 0.524
	LightGBM	Oxford	0.997 / 0.470	0.792 / 0.493
		Severson	0.995 / 0.703	0.824 / 0.601
	Random Forest	Oxford	0.996 / 0.412	0.853 / 0.540
		Severson	0.999 / 0.845	0.824 / 0.632
	XGBoost	Oxford	0.956 / 0.610	0.817 / 0.507
		Severson	0.994 / 0.708	0.714 / 0.554

TABLE V. DELONG TEST P-VALUES FOR WITHIN-DATASET AND CROSS-CHEMISTRY COMPARISONS

Dataset	Comparison	Setting	p-value
NASA	XGB vs LGBM	Within	1.4×10^{-2}
	XGB vs RF	Within	3.34×10^{-1}
	LGBM vs RF	Within	1.70×10^{-1}
CALCE	XGB vs LGBM	Within	1.1×10^{-5}
	XGB vs RF	Within	4.6×10^{-27}
	LGBM vs RF	Within	1.4×10^{-17}
Oxford	XGB (+SOH) vs XGB (−SOH)	Ablation	7.2×10^{-51}
	LGBM (+SOH) vs LGBM (−SOH)	Ablation	2.6×10^{-64}
	RF (+SOH) vs RF (−SOH)	Ablation	6.5×10^{-36}
	XGB vs LGBM	+SOH	3.0×10^{-3}
	XGB vs RF	+SOH	9.1×10^{-6}
	LGBM vs RF	+SOH	3.2×10^{-2}
Severson	XGB (+SOH) vs XGB (−SOH)	Ablation	≈0
	LGBM (+SOH) vs LGBM (−SOH)	Ablation	≈0
	RF (+SOH) vs RF (−SOH)	Ablation	≈0
	XGB vs LGBM	+SOH	1.0×10^{-20}
	XGB vs RF	+SOH	3.7×10^{-3}
	LGBM vs RF	+SOH	5.9×10^{-3}



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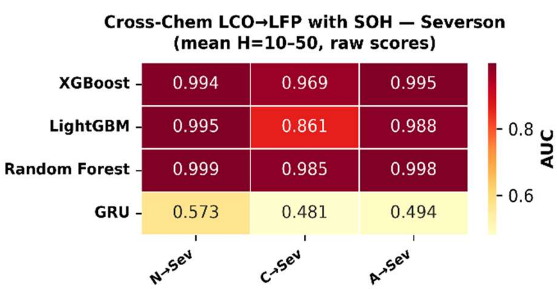

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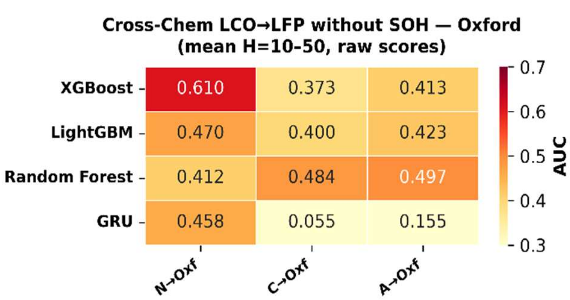
Fig. 4. Cross-chemistry raw AUC with SOH included, (a) Oxford and (b) Severson targets.



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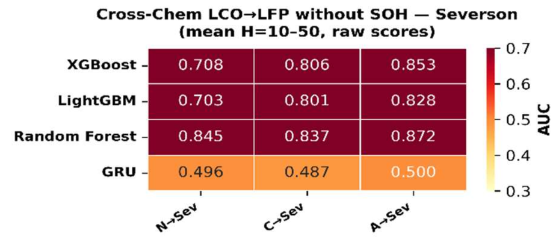

(b)

Fig. 5. Cross-chemistry raw AUC with SOH removed, (a) Oxford and (b) Severson targets.

D. Multi-Horizon Analysis

The predictive performance of the proposed framework was evaluated across four prediction horizons (H = 10, 20, 30, and 50).

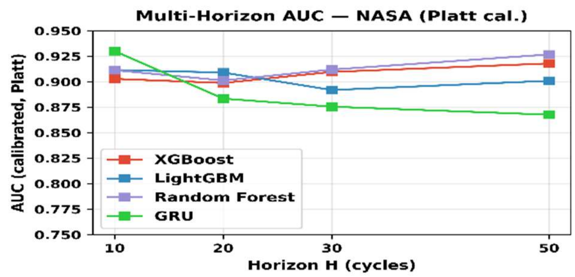

Fig. 6. Prediction performance across different operating horizons (NASA, Platt-calibrated).

As shown in Fig. 6, AUC varied by less than 0.05 across most model–dataset combinations, indicating stable performance over different prediction horizons. Slightly larger variations occurred at shorter horizons because of increased class imbalance. The Oxford dataset was excluded from multi-horizon validation because its 100-cycle sampling interval produced identical labels across all horizons. Overall, the proposed framework demonstrates consistent prediction performance across the NASA, CALCE, and Severson datasets.

V. CONCLUSION

This paper presented a multi-horizon hazard framework for predicting lithium-ion battery failure across multiple datasets and chemistries. Combining tree ensemble models (XGBoost, LightGBM, Random Forest) with a GRU baseline, probability calibration, cross-chemistry evaluation, and DeLong significance testing, the framework achieved strong within-dataset performance (AUC ≥ 0.85 in 30 of 32 model-horizon-dataset combinations), with Platt sigmoid calibration consistently outperforming isotonic regression. Cross-chemistry results showed that apparent transfer success depends primarily on SOH rather than transferable degradation patterns. Removing SOH caused a statistically significant AUC drop across all models and target datasets, confirming SOH as a chemistry-specific rather than universally transferable indicator. A few limitations are worth noting. The discrete multi-horizon formulation simplifies the continuous hazard modeling used in survival analysis [12], [18]. The framework also relies on relatively small public laboratory datasets that may not reflect real-world operating diversity, and cross-chemistry evaluation was limited to LCO and LFP, excluding NMC, NCA, and other chemistry combinations. The Oxford dataset's coarse 100-cycle sampling excluded it from the multi-horizon analysis. Calibration was fitted on training-fold predictions rather than an independent validation set, potentially yielding optimistic estimates. Finally, the framework has not been validated on embedded hardware, so its inference latency, computational overhead, and real-time deployment feasibility and thus its industrial scalability remain unestablished. Future work will extend the framework to additional chemistries and transfer scenarios, explore continuous time-to-failure modelling as an alternative to the fixed-horizon approach, and develop a hardware prototype to evaluate real-time BMS performance. Validation on large-scale industrial datasets will also be essential to confirm robustness and deployment readiness.

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